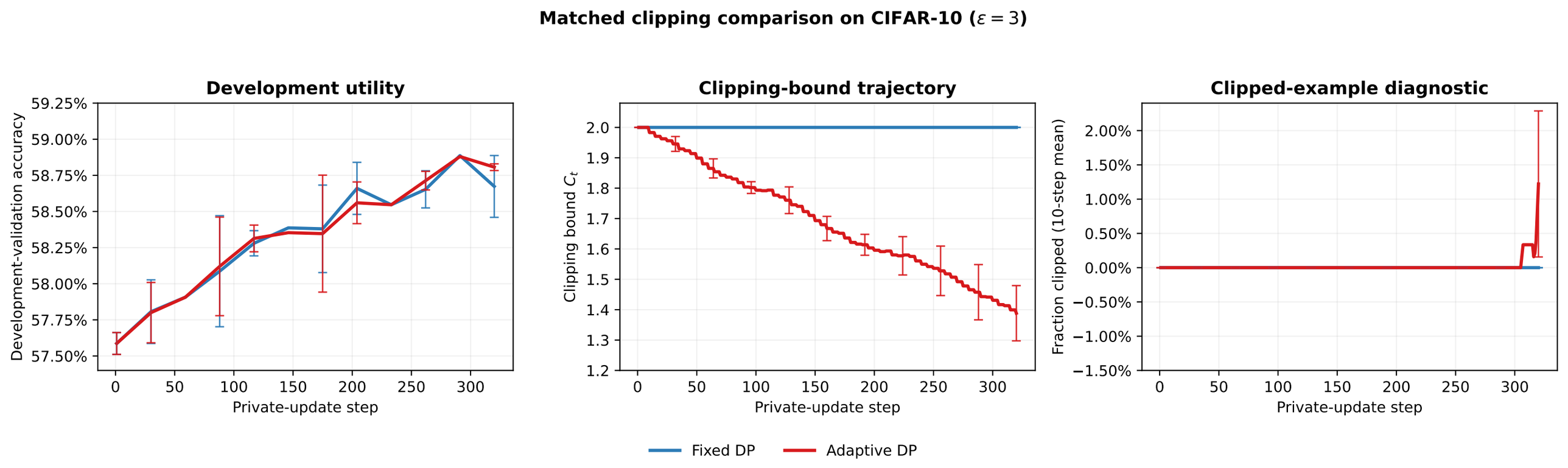
Matched clipping comparison on CIFAR-10 ($\varepsilon = 3$)
Development utility
Development-validation accuracy
59.25%
59.00%
58.75%
58.50%
58.25%
58.00%
57.75%
57.50%
0
50
100
150
200
250
300
Private-update step
Clipping-bound trajectory
Clipping bound C_t
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
0
50
100
150
200
250
300
Private-update step
Clipped-example diagnostic
Fraction clipped (10-step mean)
2.00%
1.50%
1.00%
0.50%
0.00%
−0.50%
−1.00%
−1.50%
0
50
100
150
200
250
300
Private-update step
Fixed DP
Adaptive DP
Mean with I-shaped sample-SD bars over frozen RNG seeds 42/43/44. Clipped fraction is a non-private diagnostic; official test and locked holdout were not read.